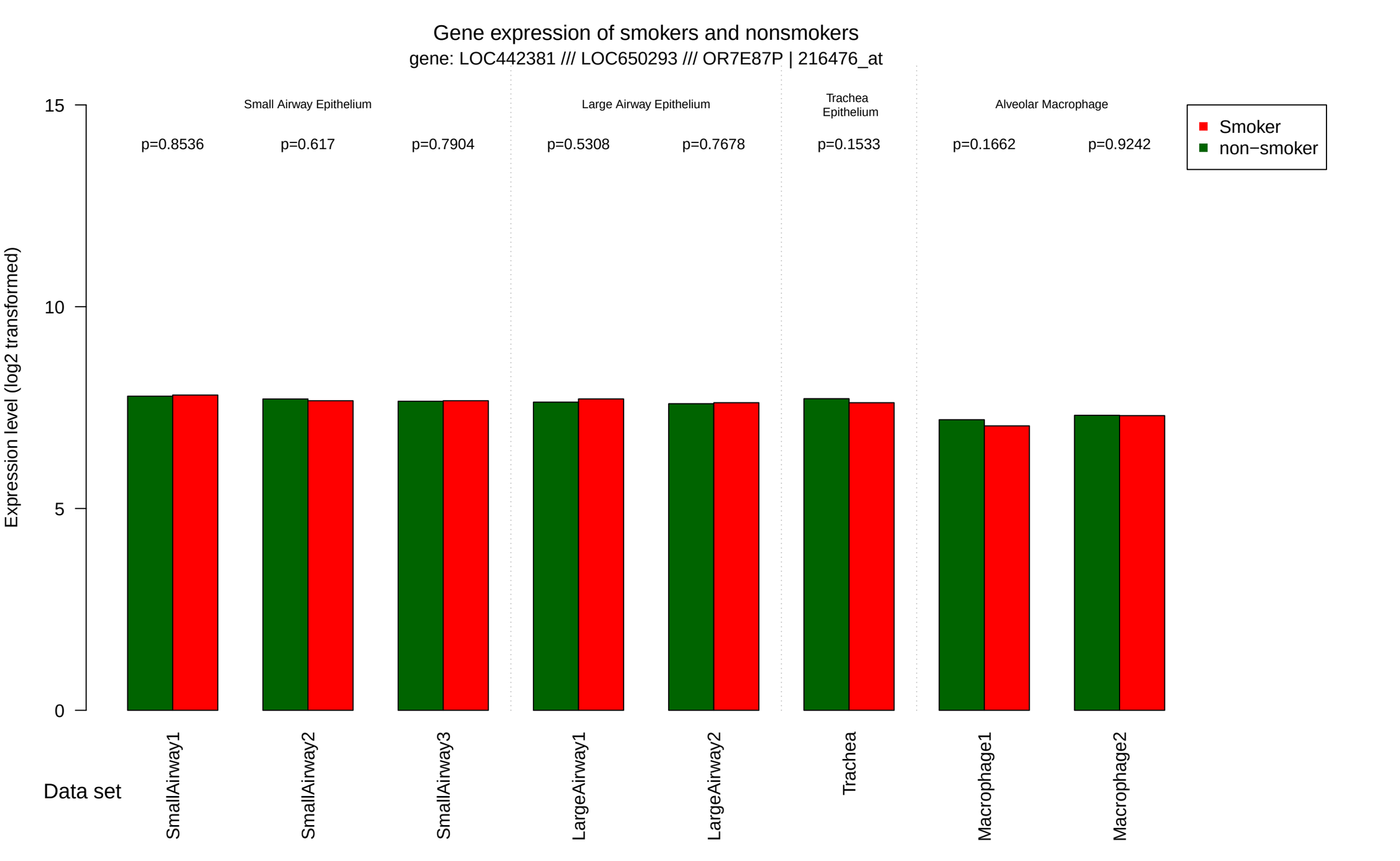
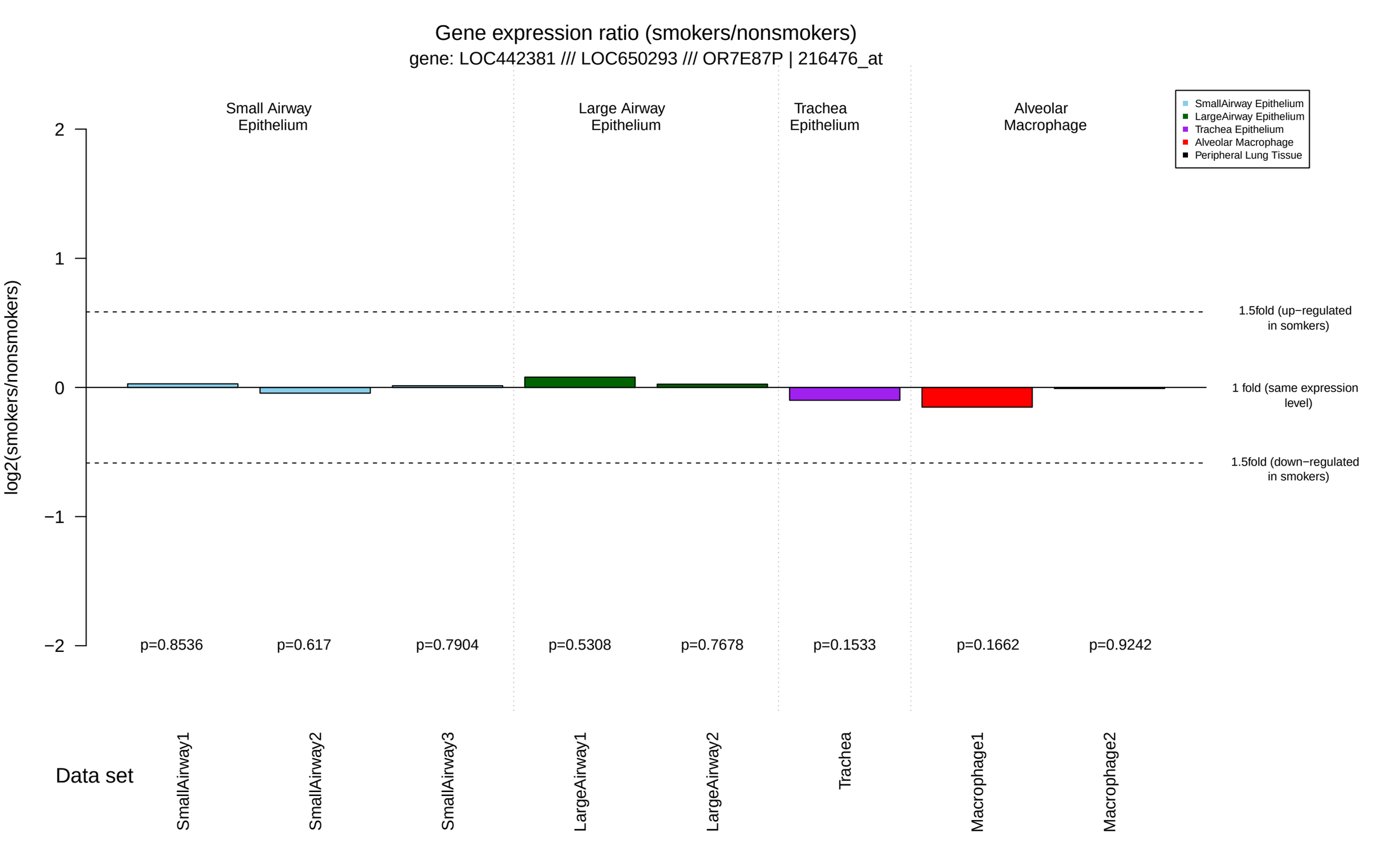
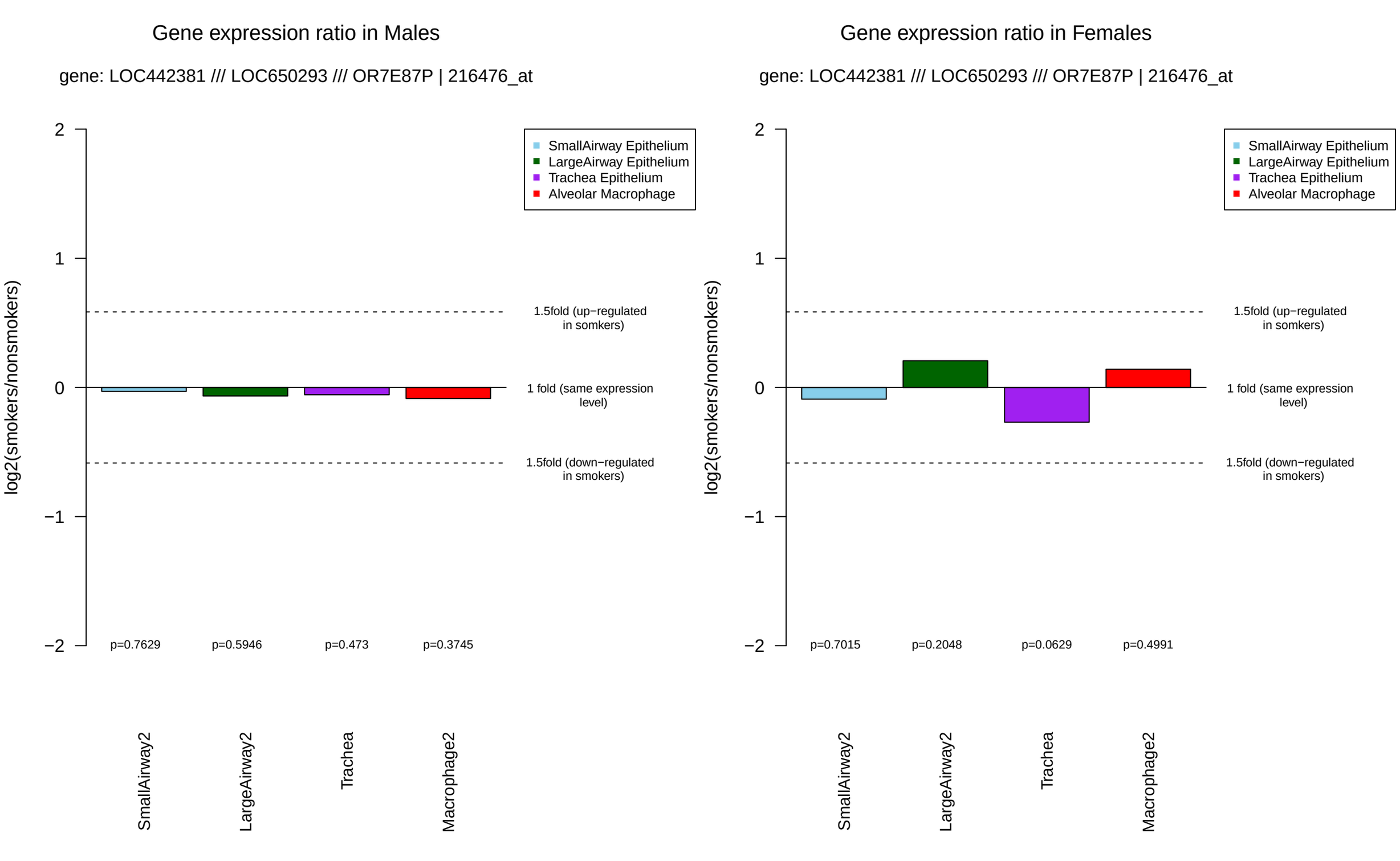
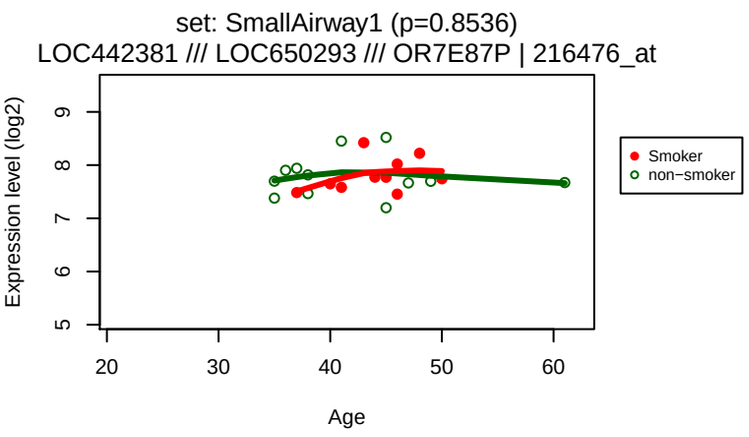
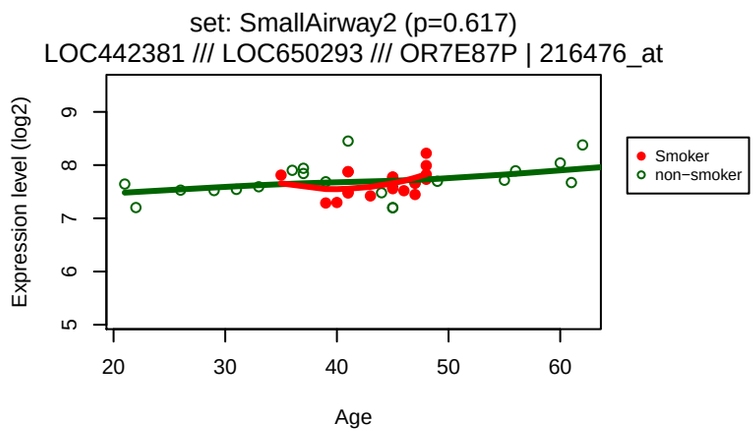
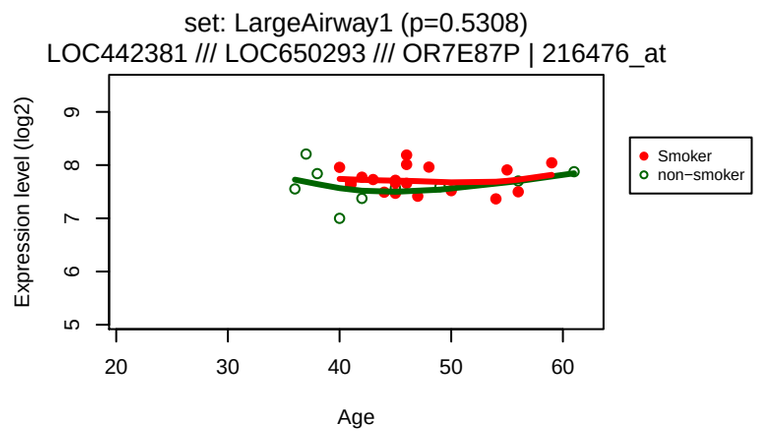

Gene expression of smokers and nonsmokers
gene: LOC442381 /// LOC650293 /// OR7E87P | 216476_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8536
p=0.617
p=0.7904
p=0.5308
p=0.7678
p=0.1533
p=0.1662
p=0.9242
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

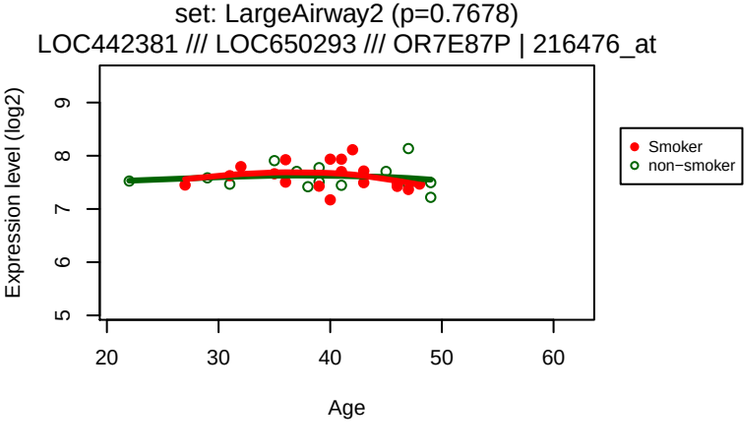

Gene expression ratio (smokers/nonsmokers)
gene: LOC442381 /// LOC650293 /// OR7E87P | 216476_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8536
p=0.617
p=0.7904
p=0.5308
p=0.7678
p=0.1533
p=0.1662
p=0.9242
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

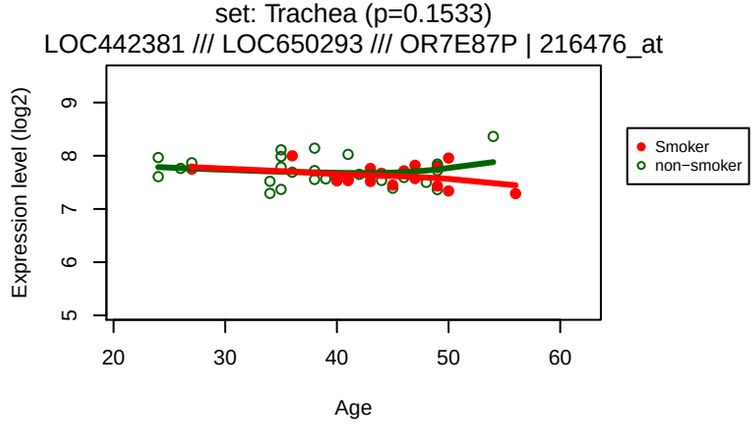

Gene expression ratio in Males
gene: LOC442381 /// LOC650293 /// OR7E87P | 216476_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7629
p=0.5946
p=0.473
p=0.3745
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC442381 /// LOC650293 /// OR7E87P | 216476_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7015
p=0.2048
p=0.0629
p=0.4991
SmallAirway2
LargeAirway2
Trachea
Macrophage2

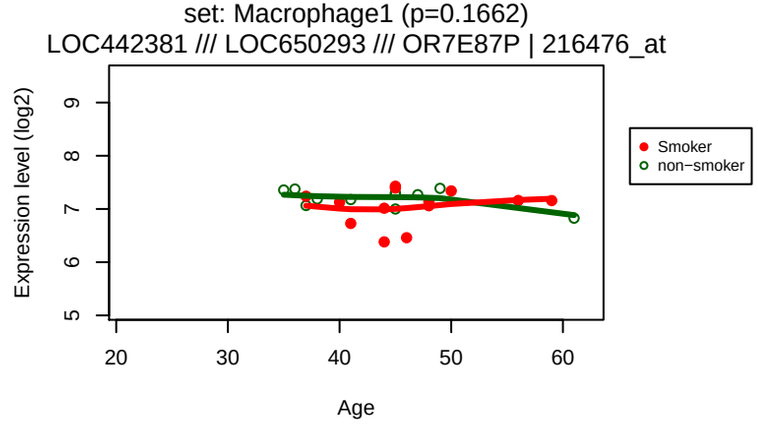

set: SmallAirway1 (p=0.8536)
LOC442381 /// LOC650293 /// OR7E87P | 216476_at
Expression level (log2)
Age
Smoker
non−smoker

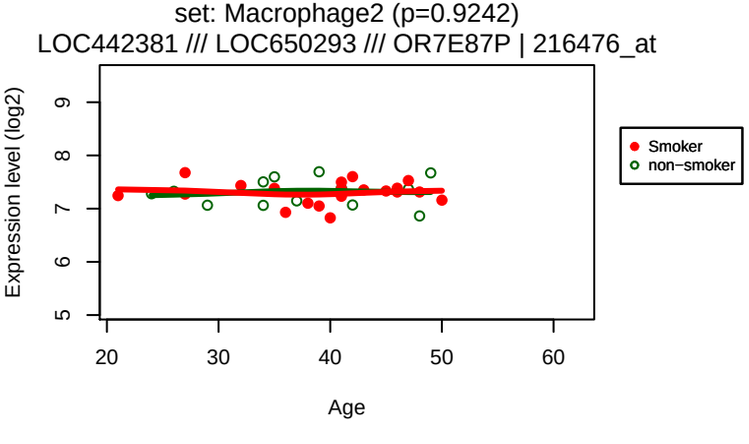

set: SmallAirway2 (p=0.617)
LOC442381 /// LOC650293 /// OR7E87P | 216476_at
Expression level (log2)
Age
Smoker
non−smoker

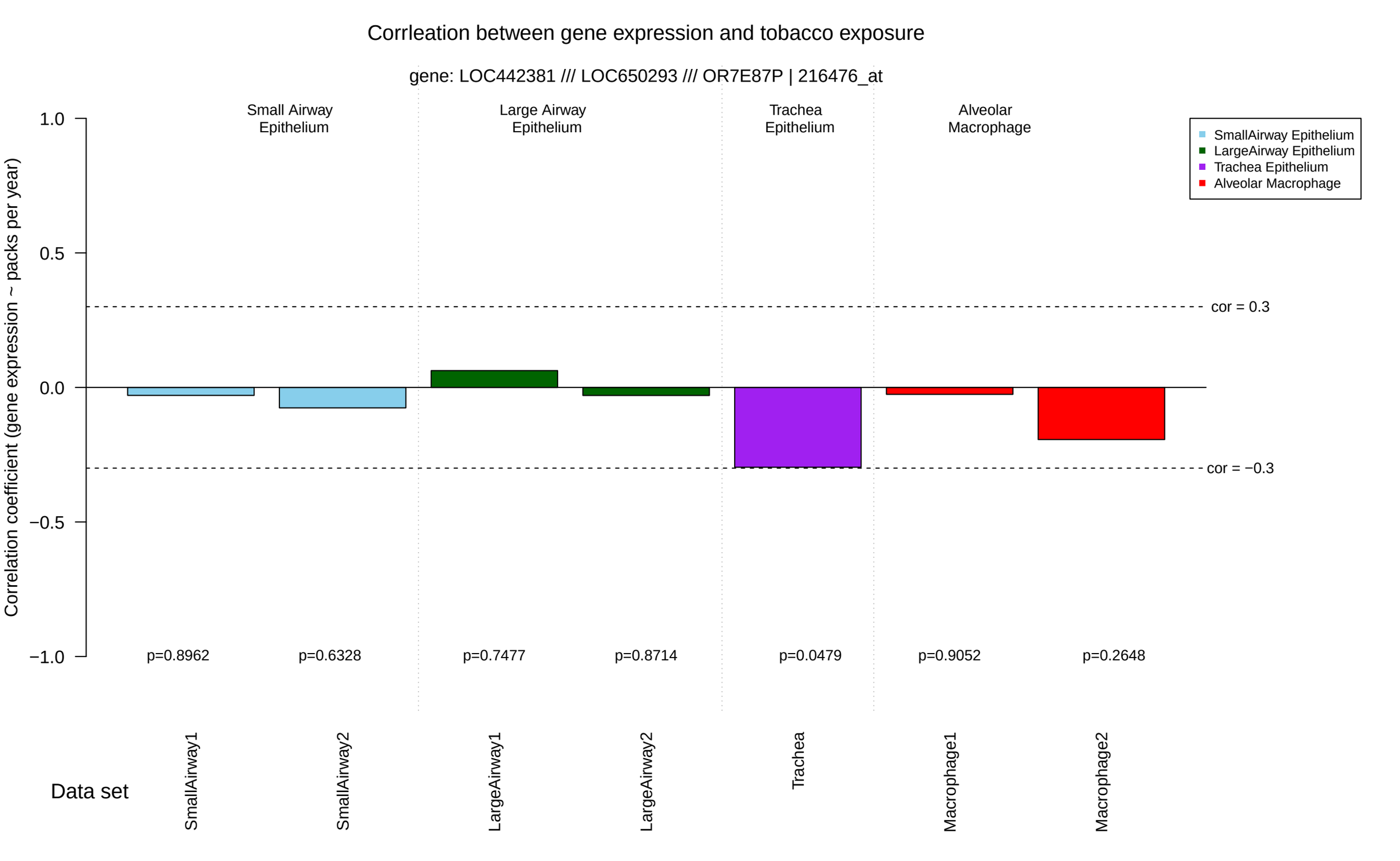

set: LargeAirway1 (p=0.5308)
LOC442381 /// LOC650293 /// OR7E87P | 216476_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7678)
LOC442381 /// LOC650293 /// OR7E87P | 216476_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1533)
LOC442381 /// LOC650293 /// OR7E87P | 216476_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1662)
LOC442381 /// LOC650293 /// OR7E87P | 216476_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9242)
LOC442381 /// LOC650293 /// OR7E87P | 216476_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC442381 /// LOC650293 /// OR7E87P | 216476_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8962
p=0.6328
p=0.7477
p=0.8714
p=0.0479
p=0.9052
p=0.2648
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2